DeviceQoSEvaluator

Interface Design Description

Eclipse Arrowhead 5.2.2 | support system | Revision 2026-11-09 | REVIEW DRAFT

Purpose and document structure

This document consolidates the system interface designs. Detailed Java service/transport IDDs are retained as separate appendices, followed by available Go and Rust interface descriptions. The current cross-language sections below precede the detailed implementation appendices. Appendix dates and release labels belong to the retained baseline, not to a newly qualified release. Existing descriptions have been assembled from the primary trees; no live interoperability or semantic-parity claim is made.

Java implementation

Primary folder: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring. 7 detailed Java IDD source document(s) are included in the appendices. Their architecture, operation parameters, models, transport bindings, access rules and errors remain associated with Java. They must not be read as automatically implemented by the ports.

Go implementation

Primary folder: Eclipse-Arrowhead-go/ah5-DeviceQoSEvaluator-go.

DeviceQoSEvaluator - Go 5.2.2

Standalone support system, reorganized on 2026-10-28 following the Eclipse-Arrowhead-java distribution layout. Build and operate from this folder; no original Arrowhead checkout or sibling source module is required. The executable is bin/DeviceQoSEvaluator. Entry point: ./cmd/deviceqosevaluator. Module: arrowhead/ah5-support-go/deviceqosevaluator. See DEPLOYMENT.md for configuration and standalone container use. Existing compatibility limits remain in the copied upstream documentation and UPSTREAM-README.md. The dedicated launcher selects only this support-system role. The formerly shared internal runtime and regression tests are copied locally, not linked from another tree. Files for common helpers and other route implementations remain in that private runtime; this packaging does not claim new functionality or Java parity. Documentation and license notices are included in this folder. Source provenance and import-only rewrites are listed in the distribution's SOURCE-MANIFEST.json. No running cloud or upstream source was changed.

Listening port

Built-in default 127.0.0.1:8472 (Arrowhead default). The shipped config.json, compose.yaml and Dockerfile use the same port on all interfaces ("listen": "0.0.0.0:8472"; 8540-8555 layout before 2026-11-09). Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Documented behavior and remaining work

System	Implemented behavior	Remaining work
DeviceQoSEvaluator	Measurement reload, weighted time-window means, threshold filtering and ranking; missing data rejected	Full Java SNMP/OID discovery, augmentation, normalization and scaling algorithm

Detailed implementation evidence

1 native Go IDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Rust implementation

Primary folder: Eclipse-Arrowhead-rust/ah5-DeviceQoSEvaluator-rust.

DeviceQoSEvaluator (Rust)

Arrowhead 5.2.2 DeviceQoSEvaluator support system. Entry point src/main.rs is built as binary deviceqosevaluator by the shared crate in ../ah5-support-runtime-rust (arrowhead-support-rust).

- examples/deviceqosevaluator.json - example configuration (master copy: ../ah5-support-runtime-rust/examples/)

- documentation/ - SysD/SysDD/IDD/SD sources (symlink into ../ah5-support-runtime-rust/documentation/DeviceQoSEvaluator, where the contract generator writes)

Run: cargo run -p arrowhead-support-rust --bin deviceqosevaluator -- <config.json>

Listening port

Default 127.0.0.1:8472 (Arrowhead default, role deviceqosevaluator). Override with "listen": "host:port" in the configuration, SERVER_PORT=N or --server.port N; the port overrides keep the configured host. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Documented behavior and remaining work

System	Implemented behavior	Remaining work
DeviceQoSEvaluator	Windowed weighted-mean filter/sort, measurement adapter	Java SNMP/OID pipeline and normalization

Detailed implementation evidence

1 native Rust IDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Build and state ownership

Core entry points are built by arrowhead-core-rust; the ordinary support entry points are built by arrowhead-support-rust using the shared support runtime. LocalCloudManager has its own crate. Per-system documentation links may point into the shared documentation catalog. Independent identity and state configuration is still required. Read the native appendix for persistence and restart semantics.

Cross-language qualification

Identical system names or HTTP paths do not establish interoperability. For the selected implementation, verify registered service versions and metadata, method/path and payload models, null and optional-field handling, success/error bodies, authentication and per-operation authorization, timeout/retry behavior and persistence across restart. Java MQTT/AMQP descriptions apply to a port only where its own implementation evidence explicitly supports them. Missing transports must not be advertised. State conversion from Java SQL to native snapshots, secret provisioning, backup/restore and rollback require implementation-specific procedures. No data migration, deployment or cloud reconfiguration was performed for this document set.

Detailed appendices and provenance

- Java: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-HTTP-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-MQTT-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-generalManagement-HTTP-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-generalManagement-MQTT-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-qualityEvaluation-HTTP-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-qualityEvaluation-MQTT-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-DeviceQoSEvaluator-java-spring/documentation/v5.2.2/DeviceQoSEvaluator-v5.2.2-sysMonitor-AMQP-IDD.tex

- Go: Eclipse-Arrowhead-go/ah5-DeviceQoSEvaluator-go/documentation/DeviceQoSEvaluator/DeviceQoSEvaluator-HTTPS-IDD.tex

- Rust: Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/DeviceQoSEvaluator/DeviceQoSEvaluator-HTTPS-IDD.tex

The package sources/ folder retains the selected LaTeX, figures, bibliography, language README/build evidence and source snapshots for newly described contracts. source-manifest.json records SHA-256 hashes of original inputs. build-report.json records whether an appendix used an existing newer PDF or was compiled from source.

deviceQualityDataManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"deviceQualityDataManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `deviceQualityDataManagement` service of the DeviceQoSEvaluator system.

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1 Overview

This document describes the `deviceQualityDataManagement` service interface of the DeviceQoSEvaluator system, which provides operator access to the stored device statistics and rebuilding of the device-system measurement map.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `deviceQualityDataManagement` service interface

This document provides the Interface Design Description IDD to the *deviceQualityDataManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `deviceQualityDataManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `device-qos-evaluator`, port 8472 and base path `/deviceqosevaluator/mgmt`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`query`	POST	`/query`
`reload`	POST	`/reload`

2 Service Operations

The interfaces of the `deviceQualityDataManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `deviceQualityDataManagement` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /deviceqosevaluator/mgmt/query

Operation: query
Input: QoSDeviceStatQueryRequest
Output: QoSDeviceStatQueryResponse

Returns a page of aggregated statistic records.

A successful call returns `200 OK` with the output in the body.

```
POST /deviceqosevaluator/mgmt/query HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "metricGroup": "RTT",
  "from": "2026-10-22T12:00:00Z",
  "to": "2026-10-22T12:00:00Z",
  "aggregation": [
    "string"
  ],
  "systemNames": [
    "TemperatureSensor"
  ]
}
```

Listing 1: An query invocation.

```
{
  "records": [
    {
      "metricGroup": "RTT",
      "id": 42,
      "timestamp": "string",
      "uuid": "string",
      "systems": [
        "string"
      ]
    }
```

```
  ],
  "count": 1
}
```

Listing 2: An query response.

2.2 POST /deviceqosevaluator/mgmt/reload

Operation: reload
Output: String

Rebuilds the device-system measurement map from the ServiceRegistry.
A successful call returns `200 OK` with the output in the body.

```
POST /deviceqosevaluator/mgmt/reload HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 3: An reload invocation.

```
"string"
```

Listing 4: An reload response.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.6, which are used to represent things like hashes, identifiers and texts.

3.1 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.2 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

3.3 struct QoSDeviceStatQueryRequest

Paged filter for device statistics.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"metricGroup"	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
"from"	DateTime	Lower bound of the entry time.
"to"	DateTime	Upper bound of the entry time.
"aggregation"	Array<String>	Statistics to return (minimum, maximum, mean, median, current).
"systemNames"	Array<String>	Systems whose devices are returned.

3.4 struct QoSDeviceStatQueryResponse

Device statistic records.

Object Field	Value Type	Description
"records"	Array<QoSDeviceStatRecord>	Statistic records of the page.
"count"	Number	Total number of matching records (all pages).

3.5 struct QoSDeviceStatRecord

Aggregated measurements of one device in one period.

Object Field	Value Type	Description
"metricGroup"	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
"id"	Number	Identifier of the record.
"timestamp"	String	End of the aggregation period.
"uuid"	String	Identifier of the measured device.
"minimum"	Object	Minimum value in the period.
"maximum"	Object	Maximum value in the period.
"mean"	Object	Mean value in the period.
"median"	Object	Median value in the period.
"current"	Object	Last measured value.
"systems"	Array<String>	Systems hosted by the device.

3.6 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “deviceQualityDataManagement – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

deviceQualityDataManagement MQTT/TLS/JSON

Interface Design Description

Service ID: *"deviceQualityDataManagement"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `deviceQualityDataManagement` service of the DeviceQoSEvaluator system.

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1 Overview

This document describes the `deviceQualityDataManagement` service interface of the DeviceQoSEvaluator system over MQTT, which provides operator access to the stored device statistics and rebuilding of the device-system measurement map.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	3.1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `deviceQualityDataManagement` service interface

This document provides the Interface Design Description IDD to the *deviceQualityDataManagement – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/deviceqosevaluator/management/`. The standalone deployment sets `mqtt.api.enabled=false`, so the MQTT interfaces are not registered there.

2 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

Operation	Request topic
`query`	`arrowhead/deviceqosevaluator/management/query`
`reload`	`arrowhead/deviceqosevaluator/management/reload`

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
}
```

Listing 2: Response envelope.

2.1 PUBLISH `arrowhead/deviceqosevaluator/management/query`

Operation: query
Input: QoSDeviceStatQueryRequest
Output: QoSDeviceStatQueryResponse

Returns a page of aggregated statistic records.

```
{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "metricGroup": "RTT",
  "from": "2026-10-22T12:00:00Z",
  "to": "2026-10-22T12:00:00Z",
  "aggregation": [
    "string"
  ],
  "systemNames": [
    "TemperatureSensor"
  ]
```

```
}
```

Listing 3: Payload of an query request.

2.2 PUBLISH `arrowhead/deviceqosevaluator/management/reload`

Operation: **reload**
Output: **String**

Rebuilds the device-system measurement map from the ServiceRegistry.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

3.1 struct MqttRequestTemplate

The request envelope.

Object Field	Value Type	Description
"traceId"	String	Identifier chosen by the requester, returned in the response.
"authentication"	String	Identity of the requester (see Section 2).
"responseTopic"	String	Topic on which the response is published.
"qosRequirement"	Number	MQTT QoS of the response (0, 1 or 2).
"params"	Object<String>	Optional parameters (the query parameters of the HTTP binding).
"payload"	Object	The operation input.

3.2 struct MqttResponseTemplate

The response envelope.

Object Field	Value Type	Description
"status"	Number	Status code.
"traceId"	String	The trace identifier of the request.
"receiver"	String	Name of the requester.
"payload"	Object	The operation output, or an ErrorMessage.

3.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.4 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

3.5 struct QoSDeviceStatQueryRequest

Paged filter for device statistics.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"metricGroup"	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
"from"	DateTime	Lower bound of the entry time.
"to"	DateTime	Upper bound of the entry time.
"aggregation"	Array<String>	Statistics to return (minimum, maximum, mean, median, current).
"systemNames"	Array<String>	Systems whose devices are returned.

3.6 struct QoSDeviceStatQueryResponse

Device statistic records.

Object Field	Value Type	Description
"records"	Array<QoSDeviceStatRecord>	Statistic records of the page.
"count"	Number	Total number of matching records (all pages).

3.7 struct QoSDeviceStatRecord

Aggregated measurements of one device in one period.

Object Field	Value Type	Description
"metricGroup"	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
"id"	Number	Identifier of the record.
"timestamp"	String	End of the aggregation period.
"uuid"	String	Identifier of the measured device.
"minimum"	Object	Minimum value in the period.
"maximum"	Object	Maximum value in the period.
"mean"	Object	Mean value in the period.
"median"	Object	Median value in the period.
"current"	Object	Last measured value.
"systems"	Array<String>	Systems hosted by the device.

3.8 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “deviceQualityDataManagement – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-SD.pdf.

[6] ——, “deviceQualityDataManagement HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-deviceQualityDataManagement-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated; missing primitive definitions added.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

generalManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"generalManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `generalManagement` service of the DeviceQoSEvaluator system.

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1 Overview

This document describes the `generalManagement` service interface of the DeviceQoSEvaluator system, which provides general management of the DeviceQoSEvaluator system: reading its log entries and its configuration.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `generalManagement` service interface

This document provides the Interface Design Description IDD to the *generalManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `generalManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `device-qos-evaluator`, port 8472 and base path `/deviceqosevaluator/general/mgmt`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`get-log`	POST	`/logs`
`get-config`	GET	`/get-config`

2 Service Operations

The interfaces of the `generalManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `generalManagement` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /deviceqosevaluator/general/mgmt/logs

Operation: get-log
Input: LogRequest
Output: LogEntryListResponse

Returns a page of the system's log entries matching the filter.

A successful call returns `200 OK` with the output in the body.

```
POST /deviceqosevaluator/general/mgmt/logs HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "from": "2026-10-22T12:00:00Z",
  "to": "2026-10-22T12:00:00Z",
  "severity": "INFO",
  "logger": "eu.arrowhead.serviceregistry.service.ServiceDiscoveryService"
}
```

Listing 1: An get-log invocation.

```
{
  "entries": [
    {
      "logId": "42",
      "entryDate": "2026-10-22T12:00:00Z",
      "logger": "eu.arrowhead.serviceregistry.service.ServiceDiscoveryService",
      "severity": "INFO",
      "message": "Service instance registered: TemperatureSensor|temperatureReading|1.0.0"
    }
  ],
  "count": 1
}
```

Listing 2: An get-log response.

2.2 GET /deviceqosevaluator/general/mgmt/get-config?keys=*{keys}*

Operation: get-config
Output: KeyValues

Returns the effective values of the requested configuration keys. Only the public keys of the system are answered; other keys, including all secrets, are ignored.

Query parameters: *keys* (List<String>, optional). List values are repeated or comma-separated.

A successful call returns `200 OK` with the output in the body.

```
GET /deviceqosevaluator/general/mgmt/get-config?keys=<keys> HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 3: An get-config invocation.

```
{
  "map": {
    "server.port": "8443"
  }
}
```

Listing 4: An get-config response.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.7, which are used to represent things like hashes, identifiers and texts.

3.1 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.2 struct KeyValues

Configuration values of the system.

Object Field	Value Type	Description
"map"	Object<String>	The configuration keys and values; secrets are not returned.

3.3 struct LogEntry

One log entry.

Object Field	Value Type	Description
"logId"	String	Identifier of the log entry.
"entryDate"	DateTime	UTC time of the log entry.
"logger"	String	Logger (Java class) that wrote the entry; as filter, its name.
"severity"	String	Log level, e.g. `INFO`, `WARN`, `ERROR`; as filter, the minimum level.
"message"	String	Text of the log entry or status message of the job.
"exception"	String	Stack trace or exception text of the log entry, if any.

3.4 struct LogEntryListResponse

List of log entries.

Object Field	Value Type	Description
"entries"	Array<LogEntry>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

3.5 struct LogRequest

Paged filter for the log entries stored in the system's database.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"from"	DateTime	Return entries from this UTC time.
"to"	DateTime	Return entries until this UTC time.
"severity"	String	Minimum log level to return.
"logger"	String	Logger (Java class) that wrote the entry; as filter, its name.

3.6 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

3.7 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “generalManagement – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

generalManagement MQTT/TLS/JSON

Interface Design Description

Service ID: *"generalManagement"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `generalManagement` service of the DeviceQoSEvaluator system.

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1 Overview

This document describes the `generalManagement` service interface of the DeviceQoSEvaluator system over MQTT, which provides general management of the DeviceQoSEvaluator system: reading its log entries and its configuration.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	3.1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `generalManagement` service interface

This document provides the Interface Design Description IDD to the *generalManagement – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/deviceqosevaluator/general/management/`. The standalone deployment sets `mqtt.api.enabled=false`, so the MQTT interfaces are not registered there.

2 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

Operation	Request topic
`get-log`	`arrowhead/deviceqosevaluator/general/management/get-log`
`get-config`	`arrowhead/deviceqosevaluator/general/management/get-config`

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
}
```

Listing 2: Response envelope.

2.1 PUBLISH `arrowhead/deviceqosevaluator/general/management/get-log`

Operation: get-log
Input: LogRequest
Output: LogEntryListResponse

Returns a page of the system's log entries matching the filter.

```
{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "from": "2026-10-22T12:00:00Z",
  "to": "2026-10-22T12:00:00Z",
  "severity": "INFO",
  "logger": "eu.arrowhead.serviceregistry.service.ServiceDiscoveryService"
}
```

Listing 3: Payload of an get-log request.

2.2 PUBLISH `arrowhead/deviceqosevaluator/general/management/get-config`

Operation: get-config
Input: Array<String>
Output: KeyValues

Returns the effective values of the requested configuration keys. Only the public keys of the system are answered; other keys, including all secrets, are ignored.

```
[
  "string"
]
```

Listing 4: Payload of an get-config request.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

3.1 struct MqttRequestTemplate

The request envelope.

Object Field	Value Type	Description
"traceId"	String	Identifier chosen by the requester, returned in the response.
"authentication"	String	Identity of the requester (see Section 2).
"responseTopic"	String	Topic on which the response is published.
"qosRequirement"	Number	MQTT QoS of the response (0, 1 or 2).
"params"	Object<String>	Optional parameters (the query parameters of the HTTP binding).
"payload"	Object	The operation input.

3.2 struct MqttResponseTemplate

The response envelope.

Object Field	Value Type	Description
"status"	Number	Status code.
"traceId"	String	The trace identifier of the request.
"receiver"	String	Name of the requester.
"payload"	Object	The operation output, or an ErrorMessage.

3.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.4 struct KeyValues

Configuration values of the system.

Object Field	Value Type	Description
"map"	Object<String>	The configuration keys and values; secrets are not returned.

3.5 struct LogEntry

One log entry.

Object Field	Value Type	Description
"logId"	String	Identifier of the log entry.
"entryDate"	DateTime	UTC time of the log entry.
"logger"	String	Logger (Java class) that wrote the entry; as filter, its name.
"severity"	String	Log level, e.g. `INFO`, `WARN`, `ERROR`; as filter, the minimum level.
"message"	String	Text of the log entry or status message of the job.
"exception"	String	Stack trace or exception text of the log entry, if any.

3.6 struct LogEntryListResponse

List of log entries.

Object Field	Value Type	Description
"entries"	Array<LogEntry>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

3.7 struct LogRequest

Paged filter for the log entries stored in the system's database.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"from"	DateTime	Return entries from this UTC time.
"to"	DateTime	Return entries until this UTC time.
"severity"	String	Minimum log level to return.
"logger"	String	Logger (Java class) that wrote the entry; as filter, its name.

3.8 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

3.9 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “generalManagement – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-SD.pdf.

[6] ——, “generalManagement HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-generalManagement-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated; missing primitive definitions added.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

qualityEvaluation HTTP/TLS/JSON

Interface Design Description

Service ID: *"qualityEvaluation"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `qualityEvaluation` service of the DeviceQoSEvaluator system.

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1 Overview

This document describes the `qualityEvaluation` service interface of the DeviceQoSEvaluator system, which provides quality-of-service evaluation of candidate providers from measurements of the devices hosting them (round-trip time, CPU load, memory use, network load).

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `qualityEvaluation` service interface

This document provides the Interface Design Description IDD to the *qualityEvaluation – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `qualityEvaluation` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `device-qos-evaluator`, port 8472 and base path `/deviceqosevaluator/qualityevaluation`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`filter`	POST	`/filter`
`sort`	POST	`/sort`

2 Service Operations

The interfaces of the `qualityEvaluation` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `qualityEvaluation` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /deviceqosevaluator/qualityevaluation/filter

Operation: filter
Input: QoSEvaluationRequest
Output: QoSEvaluationFilterResponse

Returns the providers that meet the threshold of the weighted metrics, and the dropped ones.

A successful call returns `200 OK` with the output in the body.

```
POST /deviceqosevaluator/qualityevaluation/filter HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "providers": [
    "TemperatureSensor"
  ]
}
```

Listing 1: An filter invocation.

```
{
  "passedProviders": [
    "string"
  ],
  "droppedProviders": [
    "string"
  ]
}
```

Listing 2: An filter response.

2.2 POST /deviceqosevaluator/qualityevaluation/sort

Operation: sort
Input: QoSEvaluationRequest
Output: QoSEvaluationSortResponse

Returns the providers ordered by the weighted metrics, best first.

A successful call returns `200 OK` with the output in the body.

```
POST /deviceqosevaluator/qualityevaluation/sort HTTP/1.1
Host: device-qos-evaluator:8472
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "providers": [
    "TemperatureSensor"
  ]
}
```

Listing 3: An sort invocation.

```
{
  "sortedProviders": [
    "string"
  ]
}
```

Listing 4: An sort response.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.5, which are used to represent things like hashes, identifiers and texts.

3.1 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.2 struct QoSEvaluationFilterResponse

Result of a filter evaluation.

Object Field	Value Type	Description
"passedProviders"	Array<String>	Providers that meet the threshold.
"droppedProviders"	Array<String>	Providers that do not meet it.
"warnings"	Object<Array<String>>	Per provider, warnings such as missing measurements.

3.3 struct QoSEvaluationRequest

Providers to evaluate and the evaluation settings.

Object Field	Value Type	Description
"providers"	Array<String>	Candidate provider systems.
"configuration"	Object	Evaluation settings: `metricNames` (e.g. `RTT_MEAN`), `metricWeights`, `timeWindow` (seconds) and `threshold`.

3.4 struct QoSEvaluationSortResponse

Result of a sort evaluation.

Object Field	Value Type	Description
"sortedProviders"	Array<String>	Providers ordered from best to worst.
"warnings"	Object<Array<String>>	Per provider, warnings such as missing measurements.

3.5 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters,

which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “qualityEvaluation – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-qualityEvaluation-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

qualityEvaluation MQTT/TLS/JSON

Interface Design Description

Service ID: *"qualityEvaluation"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `qualityEvaluation` service of the DeviceQoSEvaluator system.

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5.2 Quality Assurance . . . 8

1 Overview

This document describes the `qualityEvaluation` service interface of the DeviceQoSEvaluator system over MQTT, which provides quality-of-service evaluation of candidate providers from measurements of the devices hosting them (round-trip time, CPU load, memory use, network load).

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	3.1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `qualityEvaluation` service interface

This document provides the Interface Design Description IDD to the *qualityEvaluation – Service Description* document [5]. The operations, their semantics and the data models are the same as in the HTTP/TLS/JSON IDD [6]; this document specifies the MQTT binding.

The MQTT interface is offered only when the system runs with `mqtt.api.enabled=true`; it is then registered with the interface template `generic_mqtts` (`generic_mqtt` without TLS), the broker address and port (`mqtt.broker.address`, `mqtt.broker.port`) and the base topic `arrowhead/deviceqosevaluator/qualityevaluation/`. The standalone deployment sets `mqtt.api.enabled=false`, so the MQTT interfaces are not registered there.

2 Service Operations

The provider subscribes to one topic per operation, the base topic followed by the operation name:

Operation	Request topic
`filter`	`arrowhead/deviceqosevaluator/qualityevaluation/filter`
`sort`	`arrowhead/deviceqosevaluator/qualityevaluation/sort`

A consumer publishes a request envelope (MqttRequestTemplate) to the topic and receives the answer (MqttResponseTemplate) on its own `responseTopic`. The `authentication` field carries the requester's identity according to the provider's authentication policy: `IDENTITY-TOKEN//<token>` (outsourced or internal policy), `SYSTEM//<name>` (declared policy) or the requester's X.509 certificate (certificate policy). The `status` of the response uses the HTTP status code numbers (200, 201, 204, 400, 401, 403, 404, 408, 423, 500, 503); on error the payload is an ErrorMessage.

```
{
  "traceId": "a1b2c3",
  "authentication": "IDENTITY-TOKEN//<token>",
  "responseTopic": "arrowhead/response/TemperatureDisplay/a1b2c3",
  "qosRequirement": 1,
  "params": {},
  "payload": {}
}
```

Listing 1: Request envelope.

```
{
  "status": 200,
  "traceId": "a1b2c3",
  "receiver": "TemperatureDisplay",
  "payload": {}
}
```

Listing 2: Response envelope.

2.1 PUBLISH `arrowhead/deviceqosevaluator/qualityevaluation/filter`

Operation: filter
Output: QoSEvaluationFilterResponse

Returns the providers that meet the threshold of the weighted metrics, and the dropped ones.

2.2 PUBLISH `arrowhead/deviceqosevaluator/qualityevaluation/sort`

Operation: sort
Output: QoSEvaluationSortResponse

Returns the providers ordered by the weighted metrics, best first.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Each *struct* denotes a JSON Object with the named fields; fields that are not needed may be omitted.

3.1 struct MqttRequestTemplate

The request envelope.

Object Field	Value Type	Description
"traceId"	String	Identifier chosen by the requester, returned in the response.
"authentication"	String	Identity of the requester (see Section 2).
"responseTopic"	String	Topic on which the response is published.
"qosRequirement"	Number	MQTT QoS of the response (0, 1 or 2).
"params"	Object<String>	Optional parameters (the query parameters of the HTTP binding).
"payload"	Object	The operation input.

3.2 struct MqttResponseTemplate

The response envelope.

Object Field	Value Type	Description
"status"	Number	Status code.
"traceId"	String	The trace identifier of the request.
"receiver"	String	Name of the requester.
"payload"	Object	The operation output, or an ErrorMessage.

3.3 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.4 struct QoSEvaluationFilterResponse

Result of a filter evaluation.

Object Field	Value Type	Description
"passedProviders"	Array<String>	Providers that meet the threshold.
"droppedProviders"	Array<String>	Providers that do not meet it.
"warnings"	Object<Array<String>>	Per provider, warnings such as missing measurements.

3.5 struct QoSEvaluationSortResponse

Result of a sort evaluation.

Object Field	Value Type	Description
"sortedProviders"	Array<String>	Providers ordered from best to worst.
"warnings"	Object<Array<String>>	Per provider, warnings such as missing measurements.

3.6 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] A. Banks and R. Gupta, “Mqtt version 3.1.1,” OASIS, Standard, 2014.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “qualityEvaluation – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-qualityEvaluation-SD.pdf.

[6] ——, “qualityEvaluation HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-qualityEvaluation-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated; missing primitive definitions added.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

sysMonitor AMQP/TLS/JSON

Interface Design Description

Service ID: *"sysMonitor"*

Abstract
This document describes the AMQP 0-9-1 protocol with TLS payload security and JSON payload encoding variant of the `sysMonitor` service of the DeviceQoSEvaluator system (SysMonitor-AMQP-IDD-5.2.2).

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1 Overview

This document describes the AMQP binding of the native `sysMonitor` service of DeviceQoSEvaluator.
It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	AMQP	0-9-1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `sysMonitor` service interface

This document provides the Interface Design Description IDD to the *sysMonitor – Service Description* document [5].

1.1 Interface registration

With `sysmonitor.enabled=true`, DeviceQoSEvaluator registers the service at start-up with interface template `generic_amqps` (protocol `amqps`, policy `CERT_AUTH`; `generic_amqp` without TLS) and the following properties and metadata:

```
{
  "serviceDefinitionName": "sysMonitor",
  "version": "1.0.0",
  "metadata": {
    "contract": "SysMonitor-AMQP-IDD-5.2.2",
    "schemaVersion": "1.0",
    "publisherMode": "native",
    "observationTypes": [
      "echo",
      "power",
      "status"
    ],
    "delivery": "persistent, publisher-confirmed"
  },
  "interfaces": [
    {
      "templateName": "generic_amqps",
      "protocol": "amqps",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
          "amqp-broker"
        ],
        "accessPort": 5671,
        "exchange": "arrowhead",
        "baseRoutingKey": "TestCloud.sysmonitor.DeviceQoSEvaluator.DeviceQoSEvaluator",
        "operations": [
          "echo",
          "power",
          "status"
        ]
      }
    }
  ]
}
```

Listing 1: Registered service instance (TestCloud).

2 Service Operations

The publisher is the `SysMonitorPublisher` of the common library, configured with `sysmonitor.broker.address`, `sysmonitor.broker.port`, `sysmonitor.local.cloud` (default: from the system certificate), `sysmonitor.publish.interval` and `sysmonitor.ssl.*` (default: the `server.ssl` stores). The publisher connects to the broker (default `amqp-broker`, port 5671) with TLS, verifies the broker host name and authenticates with its X.509 system certificate (SASL `EXTERNAL`; the certificate CN is the broker user name). Every publication interval it publishes each observation to the topic exchange `arrowhead` with the routing key `<baseRoutingKey>.<operation>`, where the base routing key is `<Cloud>.sysmonitor.<System>.<serviceInstance>`. Messages are persistent (delivery mode 2), `application/json`, carry the observation identifier as message id, and are publisher-confirmed (timeout 5 s); a failed cycle is retried with a new connection at the next interval. The broker's topic permissions allow the system to publish only below its own base routing key.

2.1 PUBLISH `<Cloud>.sysmonitor.DeviceQoSEvaluator.DeviceQoSEvaluator.echo`

Operation: echo
Output: Observation

Liveness: `status` (`alive` or `degraded`), `uptimeSeconds` and a `detail` text; quality `good` or `uncertain`.

2.2 PUBLISH `<Cloud>.sysmonitor.DeviceQoSEvaluator.DeviceQoSEvaluator.power`

Operation: power
Output: Observation

Power supply: `supplyType=line`, `state=available`; unit `1`.

2.3 PUBLISH `<Cloud>.sysmonitor.DeviceQoSEvaluator.DeviceQoSEvaluator.status`

Operation: status
Output: Observation

Process state: `state=running`, `health` (`healthy`/`degraded`) and `serviceCount`, the number of registered services.

```
{
  "schemaVersion": "1.0",
  "observationId": "DeviceQoSEvaluator-echo-1792670400000-42",
  "providerSystem": "DeviceQoSEvaluator",
  "serviceInstance": "DeviceQoSEvaluator",
  "observationType": "echo",
  "sourceTime": "2026-10-22T12:00:00.000Z",
  "sequence": 42,
  "value": {
    "status": "alive",
    "uptimeSeconds": 3600,
    "detail": "DeviceQoSEvaluator is running"
  },
  "quality": "good",
  "metadata": {
    "publisherMode": "native"
```

```
17     }
18 }
```

Listing 2: An echo observation.

3 Data Models

3.1 struct Observation

Object Field	Value Type	Description
"schemaVersion"	String	`1.0`.
"observationId"	String	`<System>-<type>-<epoch ms>-<sequence>`.
"providerSystem"	String	The publishing system.
"serviceInstance"	String	Service instance, default the system name.
"observationType"	String	`echo`, `power`, `status`.
"sourceTime"	String	ISO 8601 UTC time with milliseconds.
"sequence"	Number	Sequence number per publisher.
"value"	Object	Observed value, see the operations.
"unit"	String	`1` for `power` and `status`; absent for `echo`.
"quality"	String	`good` or `uncertain`.
"metadata"	Object	`{"publisherMode": "native"}`.

3.2 Primitives

Type	Description
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] AMQP Working Group, “Amqp 0-9-1 specification,” AMQP Working Group, Tech. Rep., 2008.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “sysMonitor – Service Description (DeviceQoSEvaluator),” Eclipse Arrowhead, SD 5.2.2, 2026, deviceQoSEvaluator-v5.2.2-sysMonitor-SD.pdf.

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2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

1 Interface overview

HTTPS, mandatory mutual TLS, Arrowhead identity-token header and JSON request/response documents. General management is sysop/whitelist protected; enrollment endpoints use approved certificate fingerprints and profile/serial checks instead of ordinary identity tokens. Error responses contain errorMessage, errorCode, origin and timestamp. Exact DTO and status/error parity is not certified.

1.1 HTTPS interface operations

The listener is HTTPS with JSON by default; raw configuration and CRL responses use their declared media types. A request body is capped at 1 MiB. The table records implemented routes, not a claim of complete behavioral conformance.

Method	Path	Access
`POST`	`/deviceqosevaluator/qualityevaluation/filter`	identity
`POST`	`/deviceqosevaluator/qualityevaluation/sort`	identity
`POST`	`/deviceqosevaluator/mgmt/query`	management
`POST`	`/deviceqosevaluator/mgmt/reload`	management
`GET`	`/deviceqosevaluator/monitor/echo`	certificate
`GET`	`/deviceqosevaluator/general/mgmt/get-config`	management
`POST`	`/deviceqosevaluator/general/mgmt/logs`	management

1.2 Important delimitations

This implementation is a development profile, not a certified interchangeable Java replacement. All systems use private JSON snapshots rather than the Java databases. MQTT application APIs, exhaustive DTO normalization and error parity, automatic data migration and production qualification remain outstanding.

- Uses explicit DeviceMeasurementAgent input. The complete Java SNMP/OID discovery, augmentation and scaling engine is not ported; scores are weighted window means with missing-data rejection.

1 DeviceQoSEvaluator

1.1 System overview

Native Rust development profile for local Java source `DeviceQoSEvaluator/`. One independently configured executable, identity, port and state file per role. No Java, Go or Python process implements the service. Some essential operations are explicitly unavailable.

1.2 Design and trust boundaries

Tokio/Hyper/Axum with rustls enforces mutual TLS. Certificate chain, validity, local CN/profile and startup-configured optional CRLs are checked. CRLs need process restart to refresh; without CRLs no revocation enforcement is claimed. Tokens are verified with Authentication and bound to the TLS identity. Business calls require ConsumerAuthorization; management requires a verified sysop or explicit broad whitelist. CA provisioning instead applies Java-style named certificate-principal rights. Outbound authorities are pinned to expected CN/profile before credentials are sent, in addition to chain, SAN and time validation.

1.3 Persistent state

Independent locked atomic JSON replaces SQL. Clone/validate/fsync/rename commits prevent partially published local transactions. Rename is the commit point; directory-sync failure is an uncertain durable commit, not a rollback. Owner-only keys/password files are separate. Network effects are not distributed transactions; interrupted starts and bridge operations require reconciliation. No automatic Java state import, clustering or deployment is delivered.

1.4 Implemented behavior

Windowed weighted-mean filter/sort, measurement adapter

1.5 Open release gates

Java SNMP/OID pipeline and normalization. All systems remain development profiles, not certified interchangeable Java replacements. See `COMPATIBILITY.md` for detailed DTO fields and restrictions; the Java sources remain the authoritative contract.

1.6 Produced and consumed services

Advertised implemented HTTPS definitions: `deviceQualityDataManagement`, `echo`, `generalManagement`, `qualityEvaluation`.

Authorities consumed as applicable: Authentication, ServiceRegistry, ConsumerAuthorization and explicitly configured pinned adapter/provider origins. Only implemented operations are registered. Service versions are 1.0.0 except Gatekeeper/GateTunnel 5.2.2. No AMQP sysMonitor.observe implementation is advertised.

1.7 HTTPS operations

JSON unless raw configuration bytes, PEM CRL or echo text. Bodies: 1 MiB; remote responses: 8 MiB. Header: `Bearer IDENTITY-TOKEN//token`. Unknown routes return 404, absent dependencies 503. Unported

operations return 501 after applicable access checks and are excluded from registration. Route presence is not semantic parity.

Method	Path	Access / status
GET	/deviceqosevaluator/general/mgmt/get-config	management
POST	/deviceqosevaluator/general/mgmt/logs	management
GET	/deviceqosevaluator/monitor/echo	certificate
POST	/deviceqosevaluator/qualityevaluation/filter	identity
POST	/deviceqosevaluator/qualityevaluation/sort	identity
POST	/deviceqosevaluator/mgmt/query	management
POST	/deviceqosevaluator/mgmt/reload	management

Error envelope: errorMessage, errorCode, exceptionType, origin and timestamp. Full Java status, DTO and validation equivalence requires differential testing.

2 References and qualification

See contracts.json for local Java SHA256 hashes/mappings, COMPATIBILITY.md for DTO details and gaps, and VERIFICATION.md for tests actually run. No upstream release-tag provenance, complete replacement or real-cloud interoperability is asserted.